

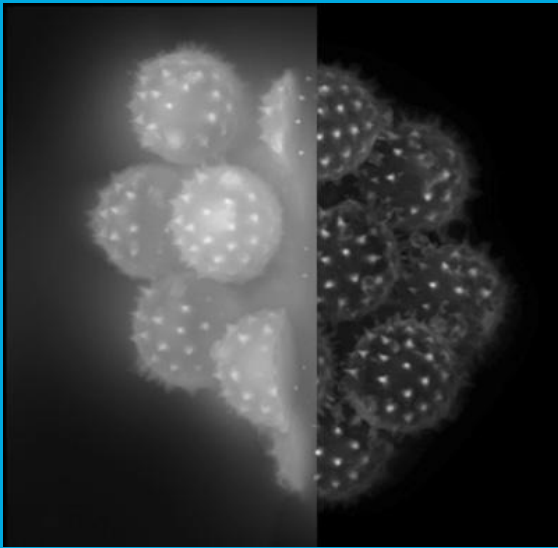


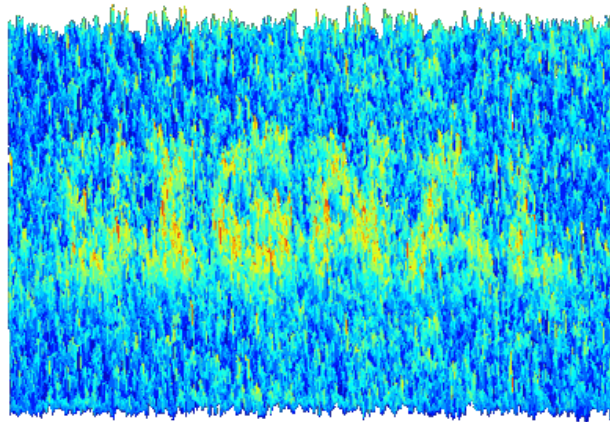
Image Degradation, Types of Noise, Image Restoration and Optimal Filtering

Lectures on Image Processing

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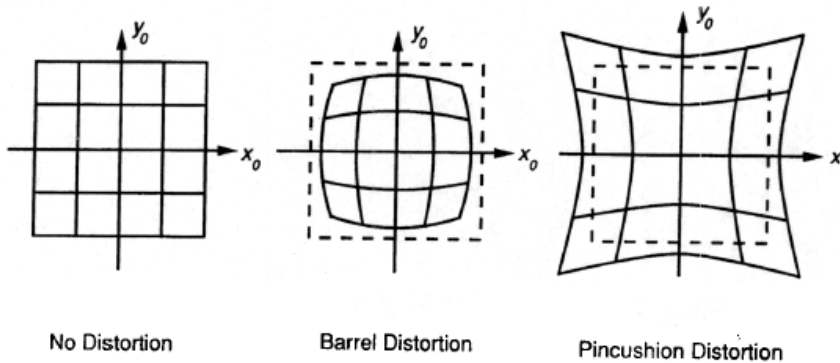


„Real image is not ideal, it is always affected by different kinds of degradations, e.g. noise“

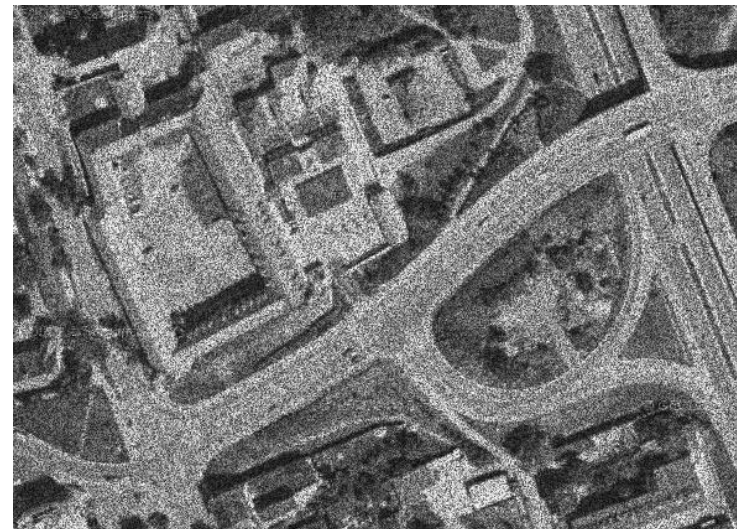


- Image degradation
- Signal-to-noise ratio
- Noise
 - Noise models
 - Image averaging
 - Local filtering
- Image restoration
 - Inverse filtering
 - Optimal filtering
 - Wiener filter

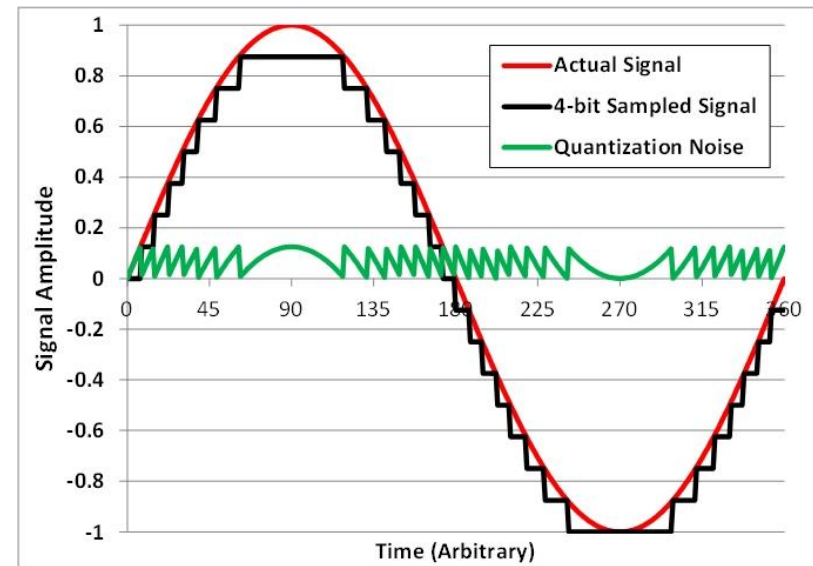
- Image distortion – e.g. geometric distortion of satellite images due to the Earth shape



- Image noise
 - Image acquisition
 - Image transmission
 - Image pre-processing



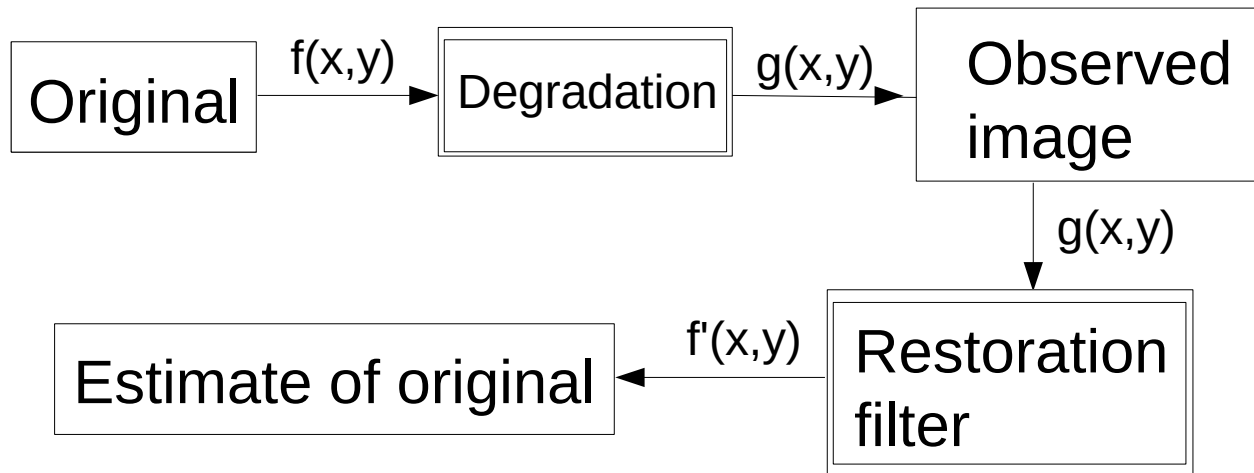
- Sensor temperature (higher temp implies noise amplification)
- Light and environmental conditions (low light conditions require high gain amplification)
- Quantization noise



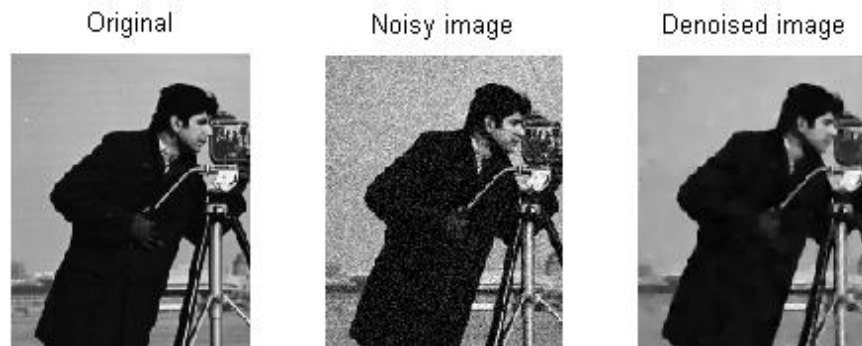
- Optical system defects (lenses)
- Relative motion between object and camera
- De-focused images



- Goal of image restoration is to improve an image for further processing



- Estimate should be as close as possible to the original!



- “Small” defect of its optical mirror (about 1/50 of the thickness of the human hair) caused the telescope to produce blurry images...



↑ Telescope's Point Spread Function – PSF

↓ Resulting blurry image



- Image restoration methods (deconvolution) can be applied as a post-processing step



↑ Observed
image

↓ Estimate



- Later fixed by corrective optics...



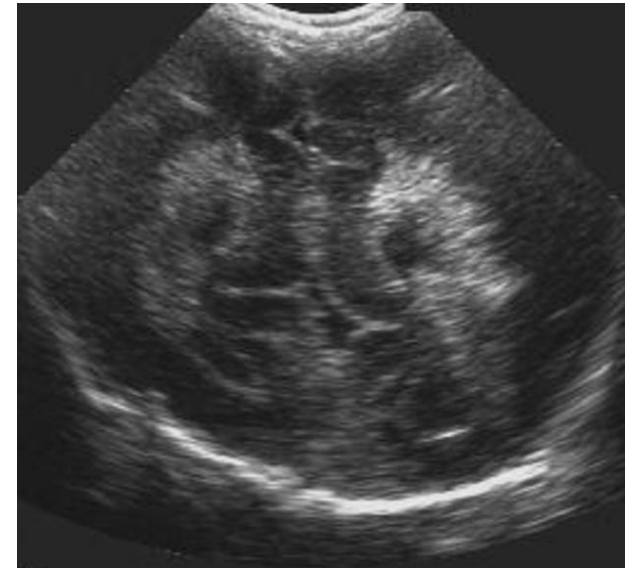
↑ Without the
corrective optics

↓ With the corrective
optics



- Independent noise – independent of the signal intensity at each pixel
- vs. signal-dependent noise (e.g. speckle noise in US images)
- Additive noise – image $f(x,y)$ and noise $v(x,y)$ are independent

$$g(x, y) = f(x, y) + v(x, y)$$



↑ Speckle noise in ultrasound image

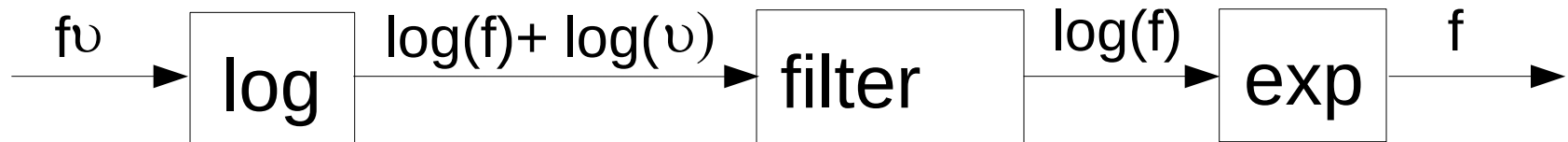
- vs. Multiplicative noise – noise intensity depends on the image intensity

$$g = f + vf = f(1 + v) \approx fv$$

- Signal-dependent (multiplicative) noise

$$g = f + \nu f = f(1 + \nu) \approx f\nu$$

- Homomorphic filtering of multiplicative noise

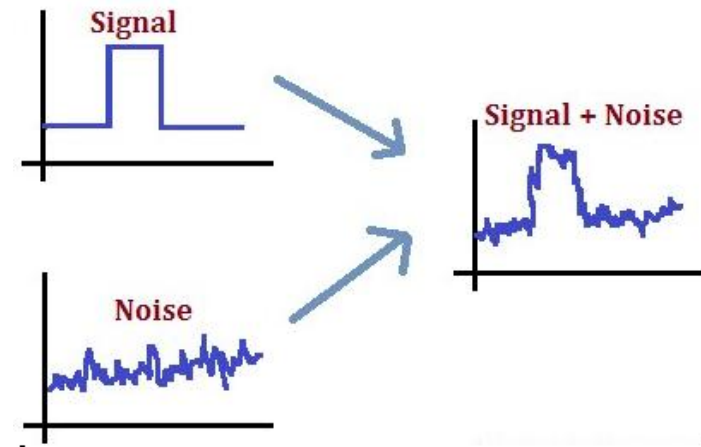


- Compares the level of a signal to the level of background noise

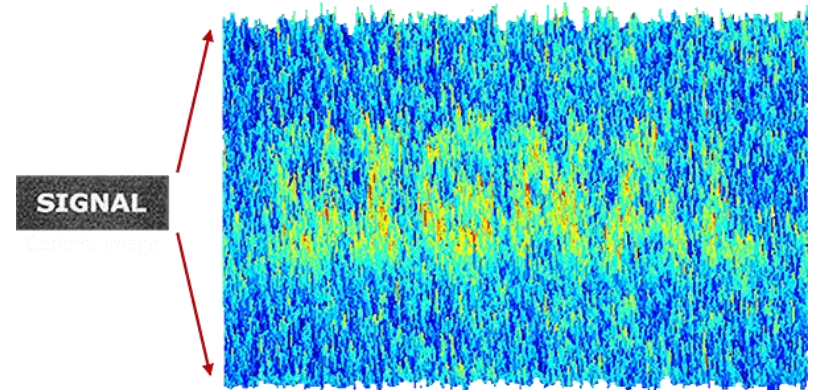
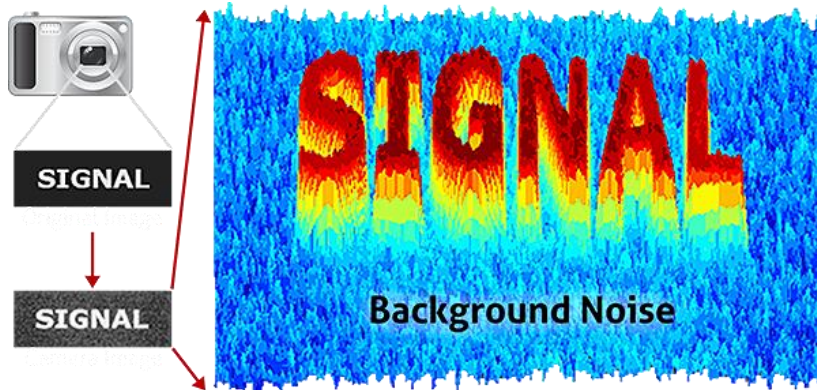
$$\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}}$$

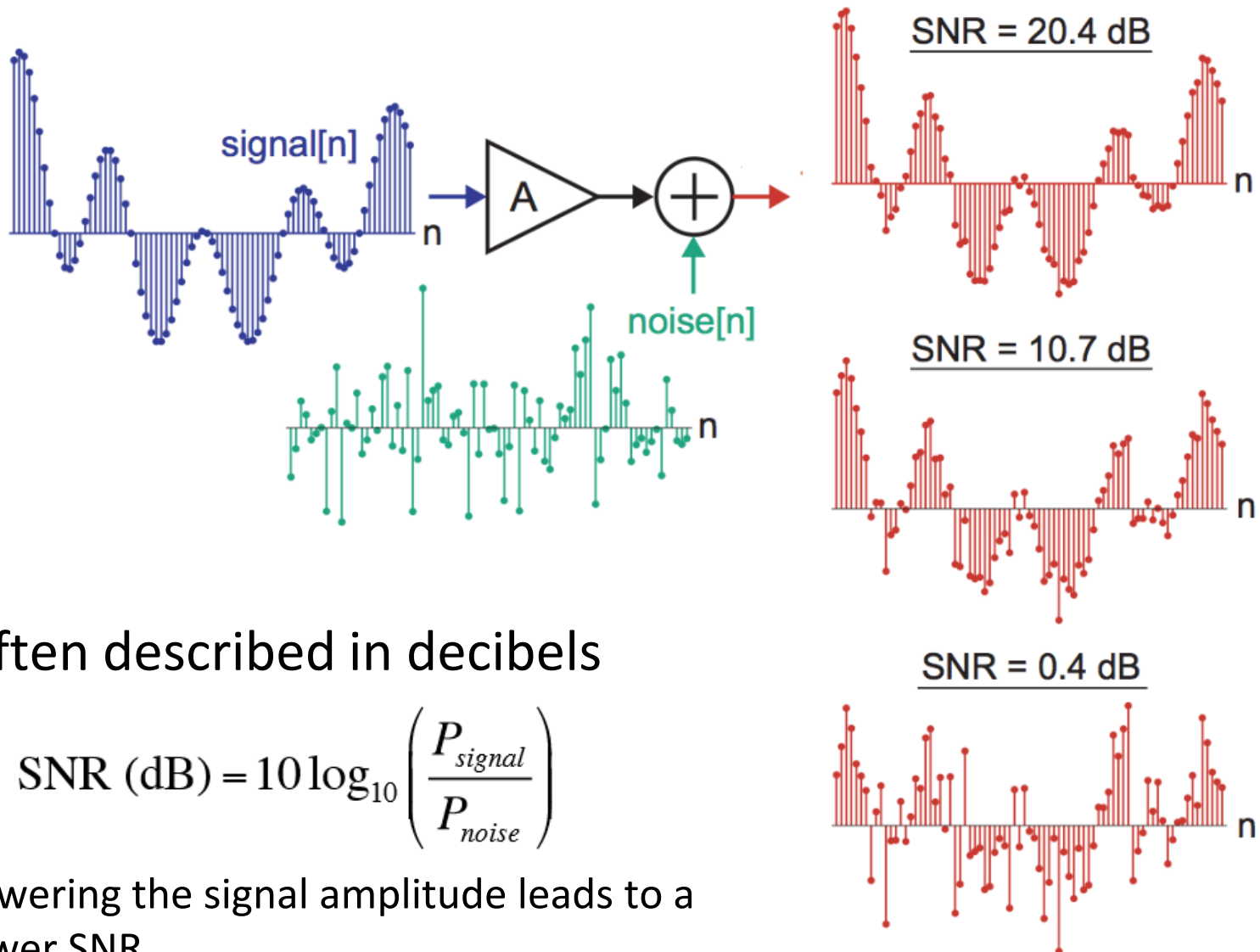
P_{signal} ... power of signal

P_{noise} ... power of noise



- Signal quality decreases for lower SNR values





- Often described in decibels

$$SNR \text{ (dB)} = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right)$$

- Lowering the signal amplitude leads to a lower SNR

$$SNR = 10 \cdot \log_{10} \left(\frac{\sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |I(i, j)|^2}{\sum_{i=0}^{m-1} \sum_{j=0}^{n-1} |I(i, j) - G(i, j)|^2} \right)$$

$I(x, y)$... original image without noise
 $G(x, y)$... image with noise

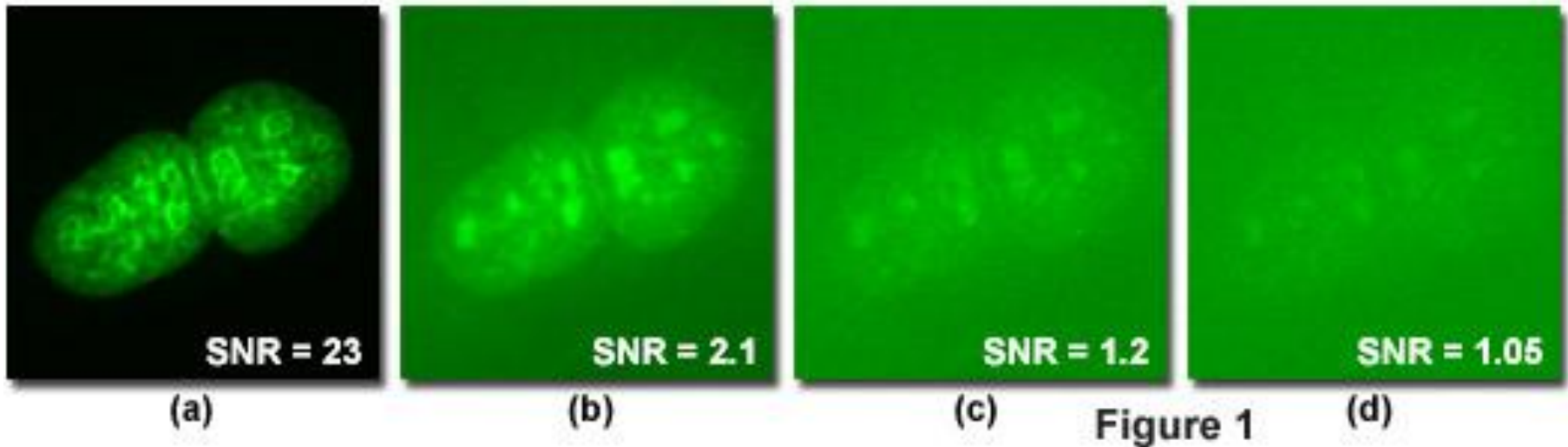
- Peak Signal-to-Noise Ratio (PSNR)
 - MAX_I ... maximum possible pixel value (e.g. 255 for 8-bit images)

$$MSE = \frac{1}{m \cdot n} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2$$

$$PSNR = 10 \cdot \log_{10} \left(\frac{MAX_I^2}{MSE} \right)$$

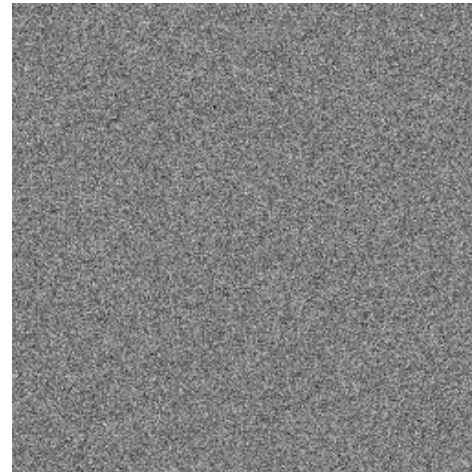
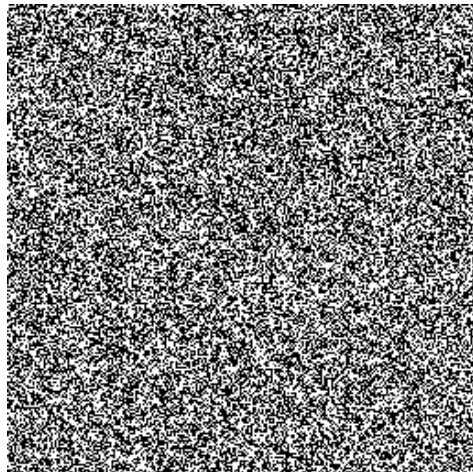
$$= 20 \cdot \log_{10} \left(\frac{MAX_I}{\sqrt{MSE}} \right)$$

Signal-to-Noise Ratios in Fluorescence Microscopy



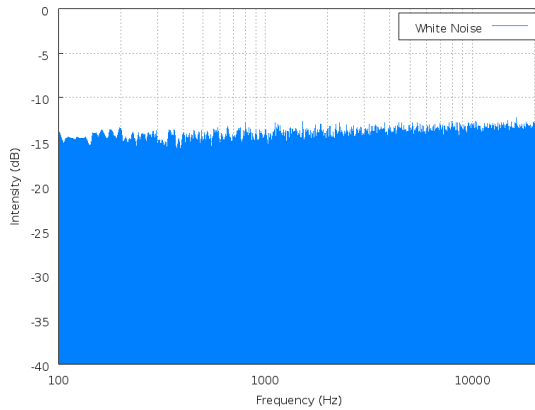
- White noise
- Gaussian noise
- Impulse “Salt-and-Pepper” noise
- Periodic noise
- ...

- Noise is uncorrelated (i.e. random)
- Noise intensity does not decrease with increasing frequency (noise has constant power spectrum)
- Approximation of the worst types of noise.

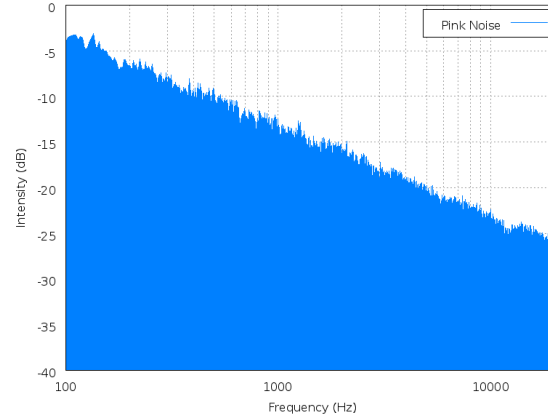


↑ White noise
examples

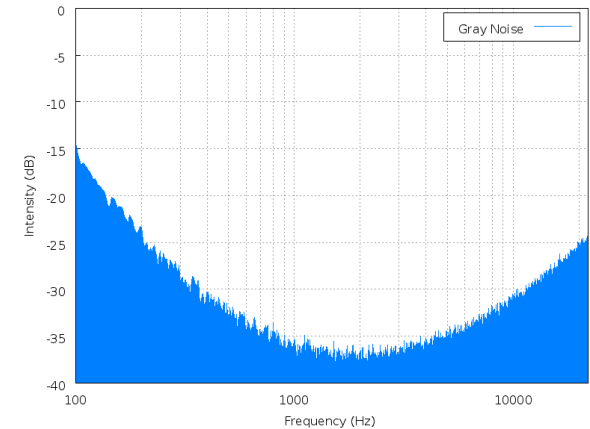
- Different assumptions about noise frequency characteristics



↑ **White noise** (flat power spectrum, affects all frequencies).



↑ **Pink noise** (frequency spectrum is linear in logarithmic scale).

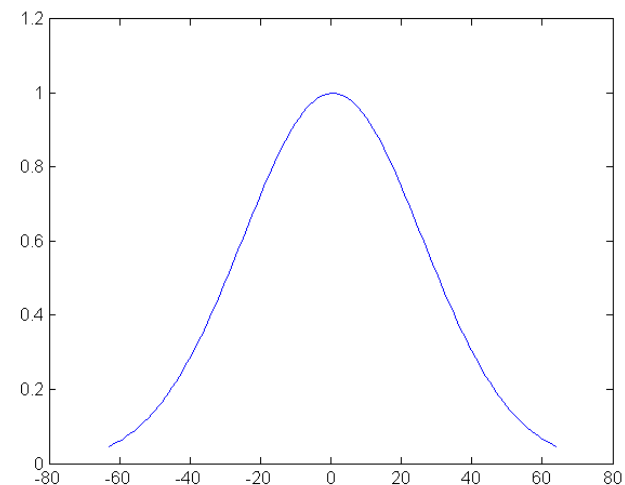


↑ **Gray noise**

- Affects all pixels of the image.
- Noise amplitude has a normal (Gaussian) probability distribution.

$$f(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right).$$

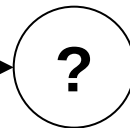
- Mean value μ (usually $\mu = 0$)
 - Standard deviation σ
- White Gaussian noise approximates real noise very well in many situations...
 - *Sensor noise caused by poor illumination and/or high temperature, electronic circuit noise, ...*



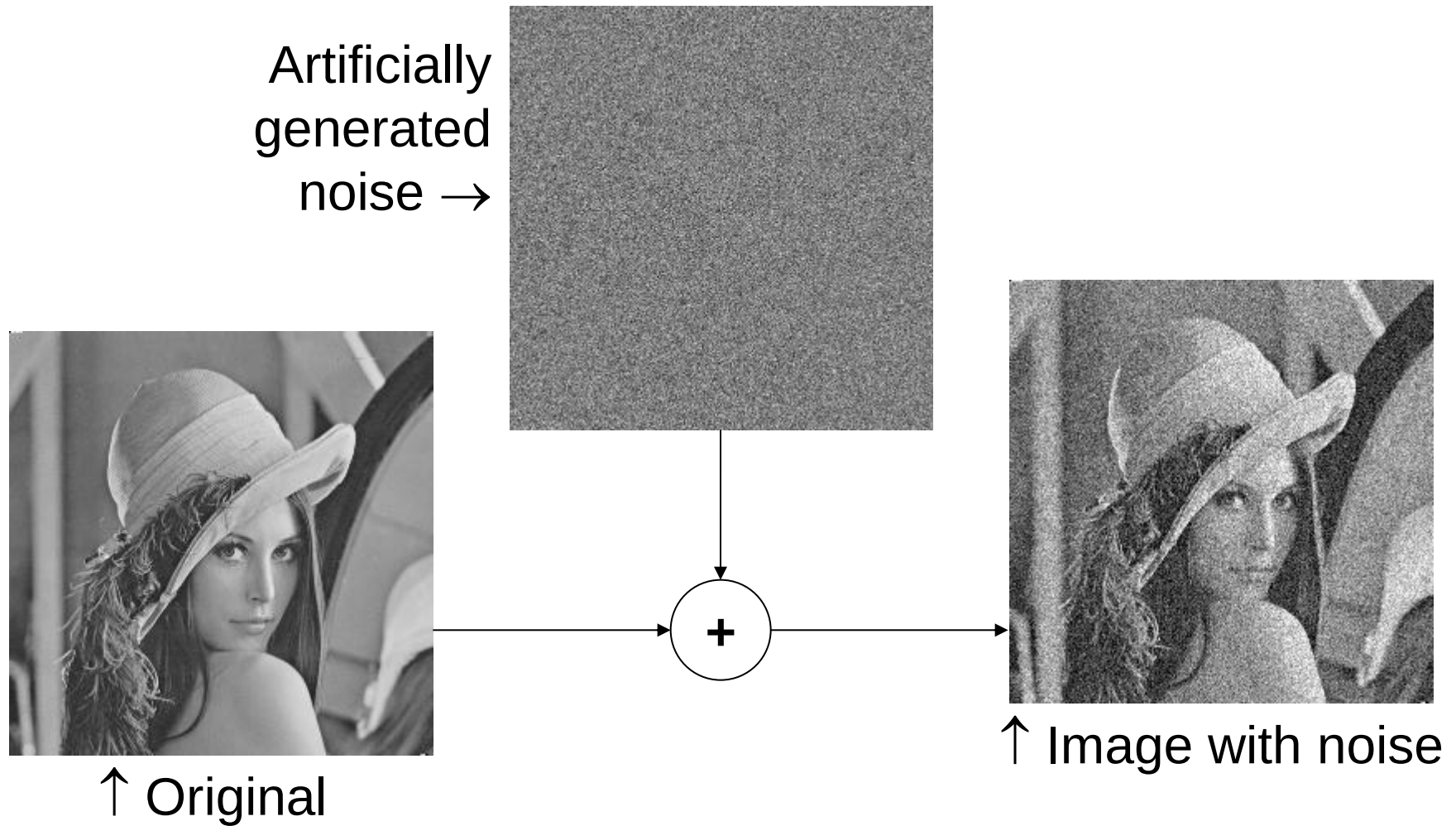
- How to artificially add Gaussian noise to an image?



↑ Original



↑ Image with noise

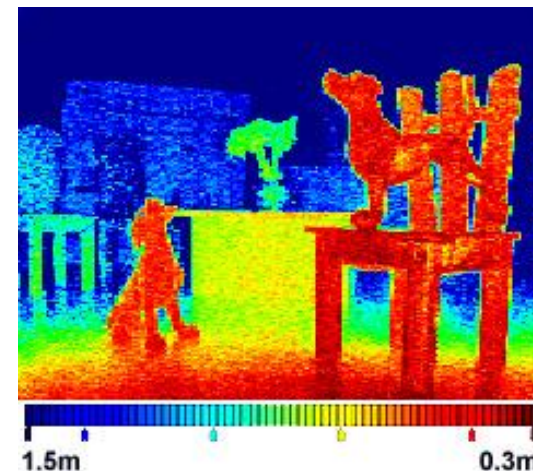
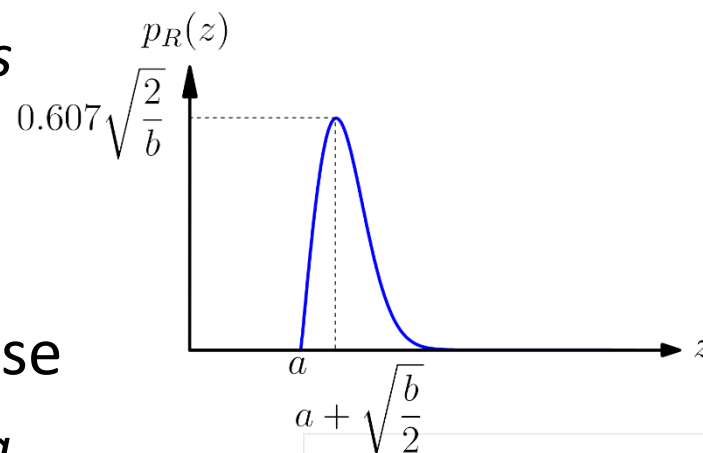


- Gaussian noise is often used in situations in which they are marginally applicable at best!

- Rayleigh noise

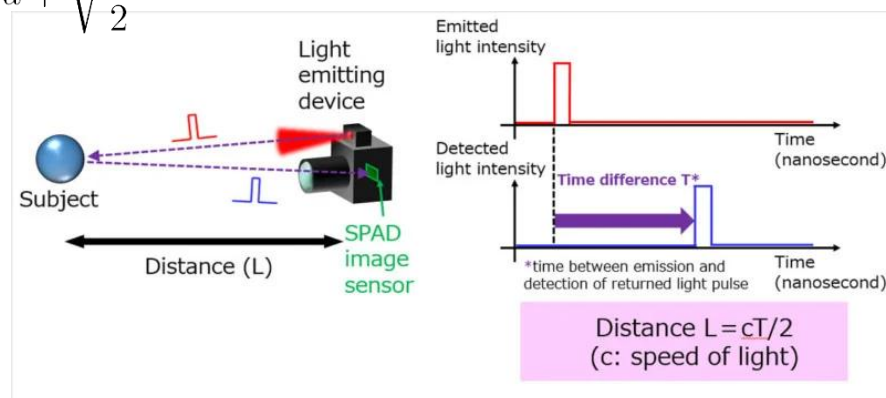
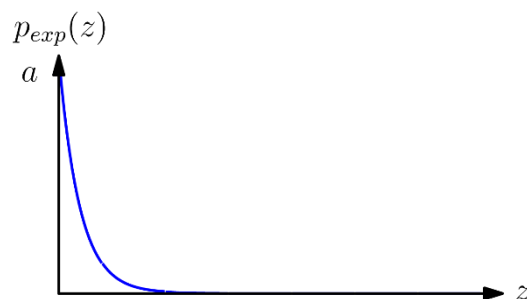
- Shape of its density function is skewed

- *Range images*

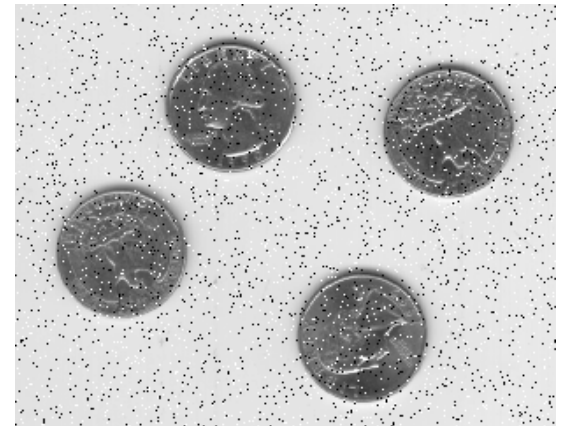
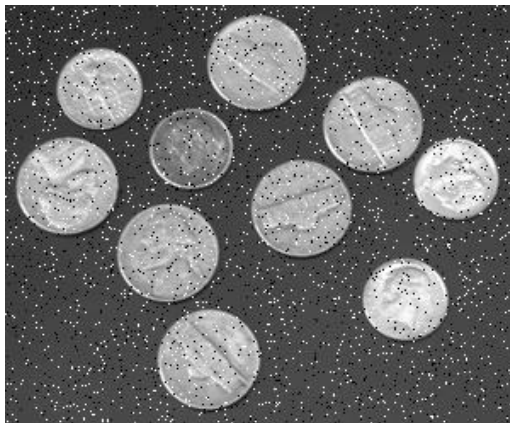


- Exponential noise

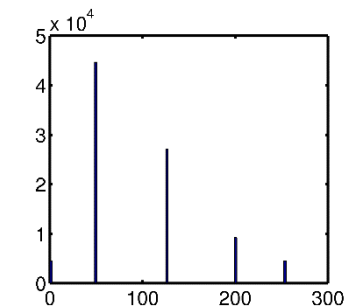
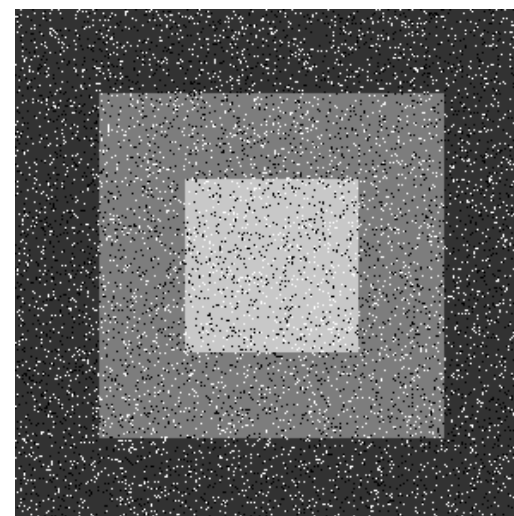
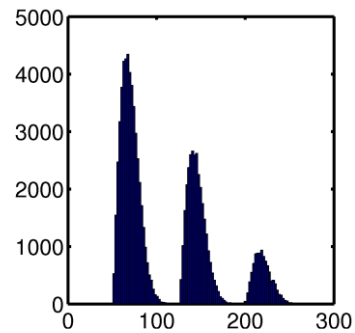
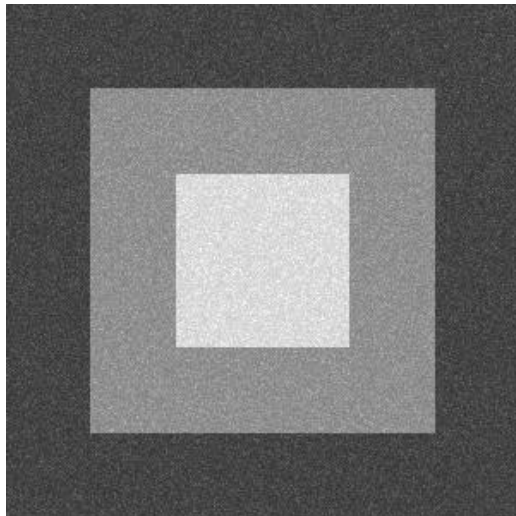
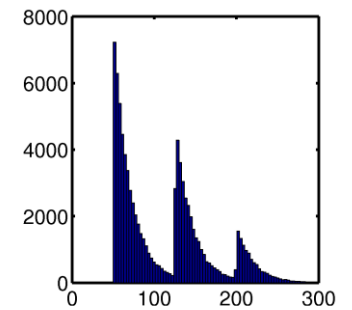
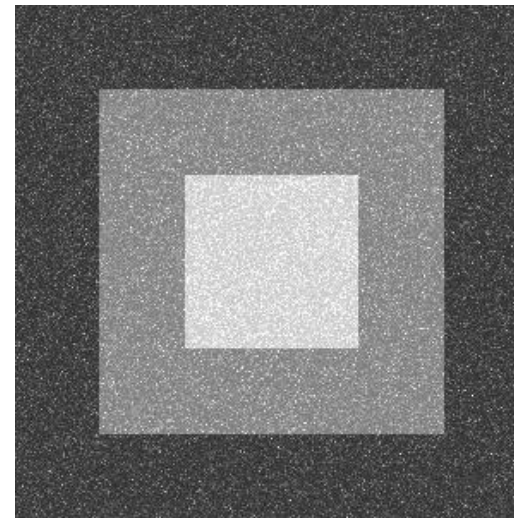
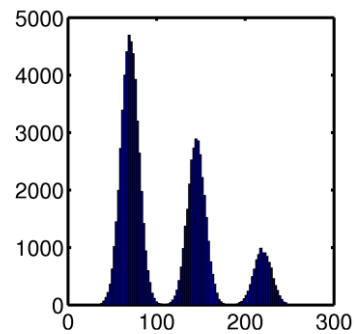
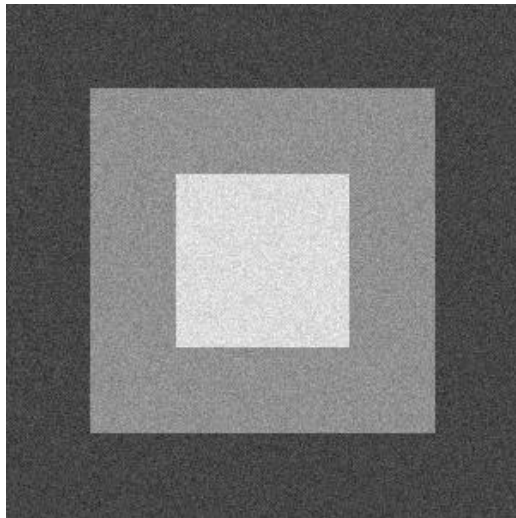
- *Laser imaging*



- Only isolated points of the image are affected
- Saturated values – black and white
 - *Faulty pixels, ...*



- Detect affected points + correct by interpolating from surrounding points

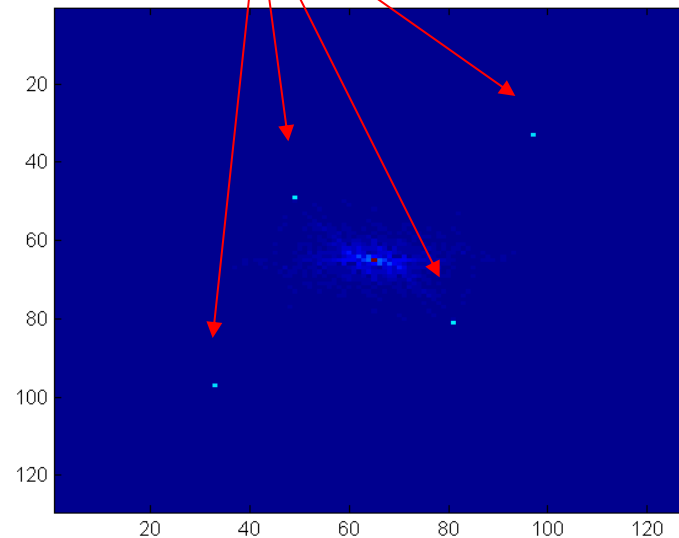


- Only some image frequencies are affected by noise
 - *Electromechanic interference during image acquisition*
- The solution is filtering in the frequency domain



↑ Image with narrow-band periodic noise

Noise → band reject filter



↑ Image spectrum (artificial case)

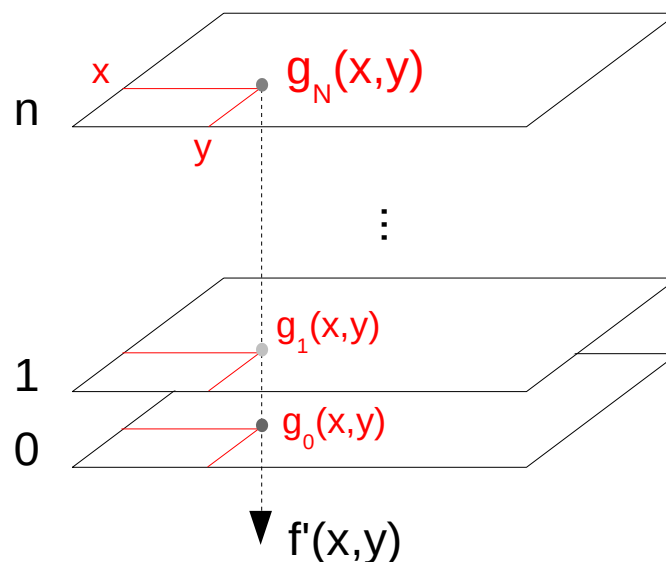


↑ After filtering and inverse FFT

- Each pixel of the image is affected with random additive noise that is
 - Signal independent
 - has zero mean value, $\mu = 0$.
 - and some standard deviation σ .
- Averaging from multiple images – averaging without blurring
- Averaging inside an image

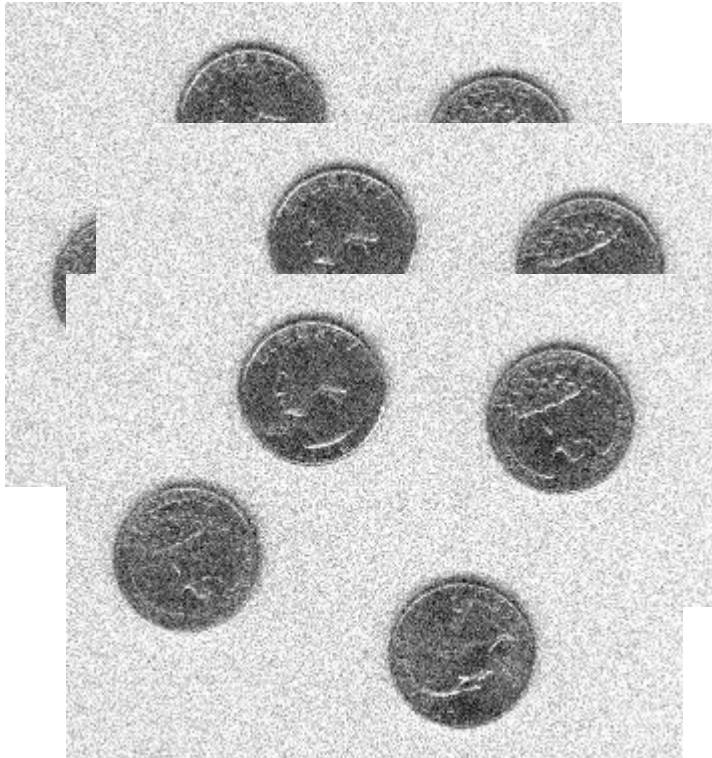
- Let's have n measurements of the **same scene**
- Correct intensity value is estimated for each pixel of the image from a set of pixels at the same position (x, y) in all measurements

$$f'(x, y) = \frac{1}{n} \sum_{i=1}^n g_k(x, y)$$

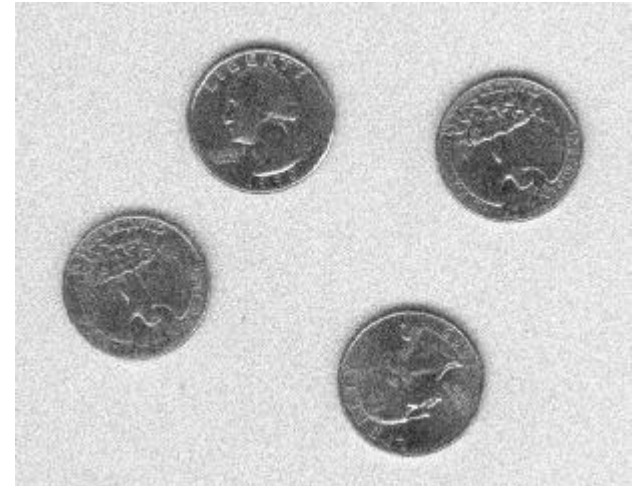


- E.g. suppression of camera thermal noise for accurate measurements (typically about 50 frames)

- Additive Gaussian noise, $\mu = 0$



Averaging of
5 images



Averaging of
50 images



- We often need to filter noise from a single image...
- Average of repeating areas – basically several “images” in one image
- Averaging inside an image
 - Neighboring pixels should have the same or similar brightness value → estimate new value (or average) from neighborhood
 - **The problem of edge blurring occurs**

- Linear filter
- Non-linear filter
- Impulse response
- Convolution
- Frequency domain filtration

- Usually, there is no specific knowledge of the image and the nature of the degradation
- The correct pixel value is estimated from a small neighborhood
- Linear filters
 - Local averaging (smoothing filters)
- Non-linear filters
 - Rotating mask method
 - Median filter
 - Bilateral filter

- Averaging in 3x3 and 5x5 neighbourhood

$$h_9 = \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$h_{25} = \frac{1}{25} \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

- Stress importance of pixels close to the center

$$h_{10} = \frac{1}{10} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$h_{10} = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$



Original



Added
gaussian
noise



After
averaging



←
Original



←
Original
+ salt-and-
pepper
noise



←
After
averaging

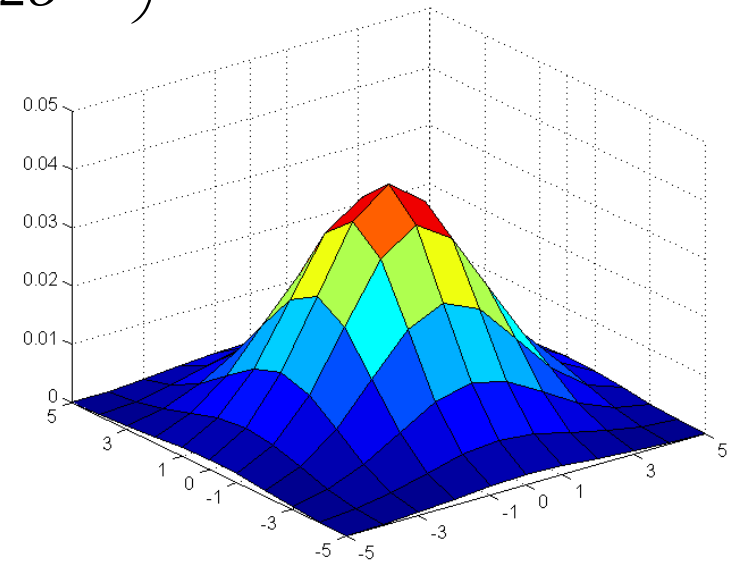
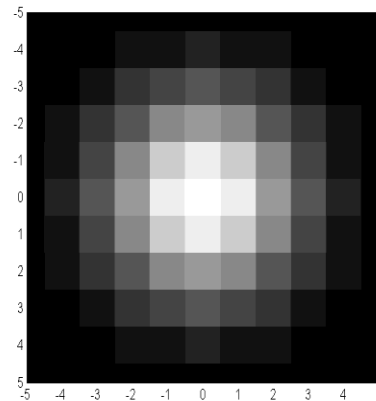
- Gaussian is a linear filter whose convolutional kernel is defined by

$$G(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right)$$

$\frac{1}{273}$

1	4	7	4	1
4	16	26	16	4
7	26	41	26	7
4	16	26	16	4
1	4	7	4	1

↑ 2D Gaussian
kernel, $\sigma = 1$



← ↑ 2D Gaussian
function, $\sigma = 2$



←
Original



←
Added artif.
Gaussian
noise

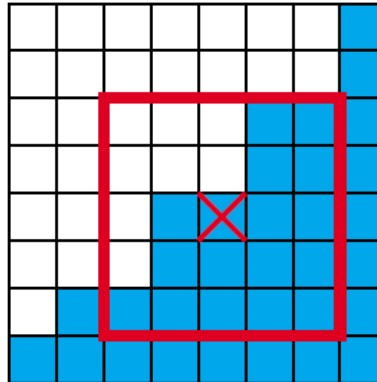


←
Gaussian
 $\sigma = 1.5$



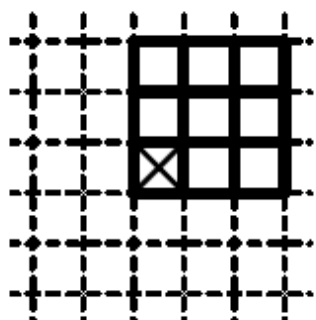
←
Gaussian
 $\sigma = 2$

- Edge blurring occurs when smoothing
- Find a subset of the pixel neighborhood, in which the intensity does not change much → *Rotating mask method*

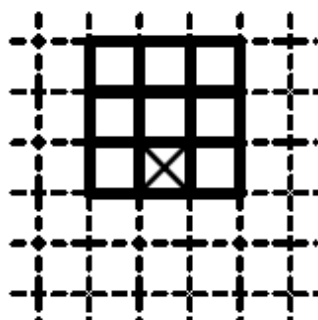


- Mean value is a bad estimate if outliers are present → *Median filter*

- In the neighborhood of 5x5, a 3x3 rotating mask looks for a homogeneous part
- There are nine possible positions...

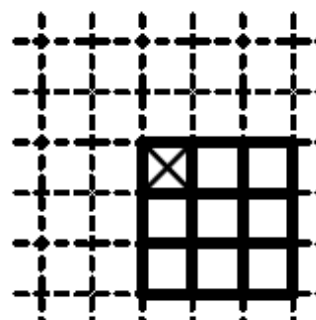


1

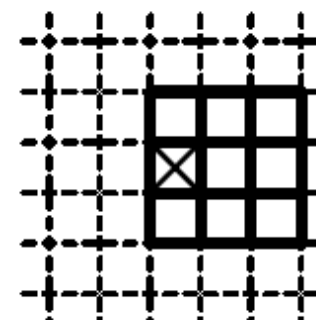


2

...

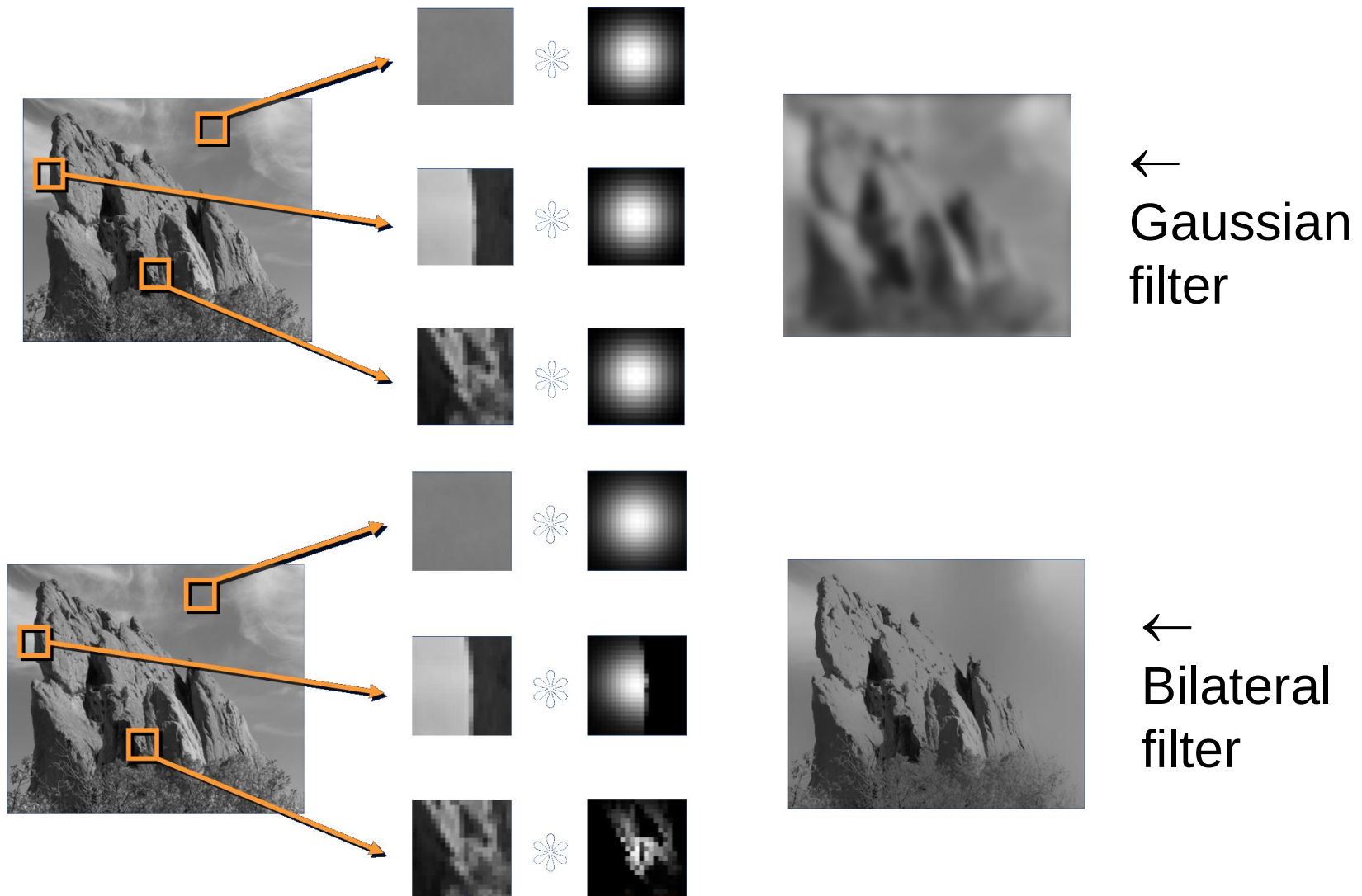


7



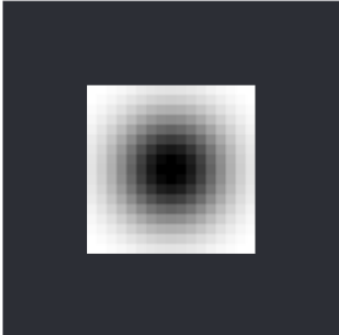
8

- Mask with the minimal dispersion of intensity is selected



- Space weight - distance from the center (Gaussian filter)
- Range weight - intensity difference from the center

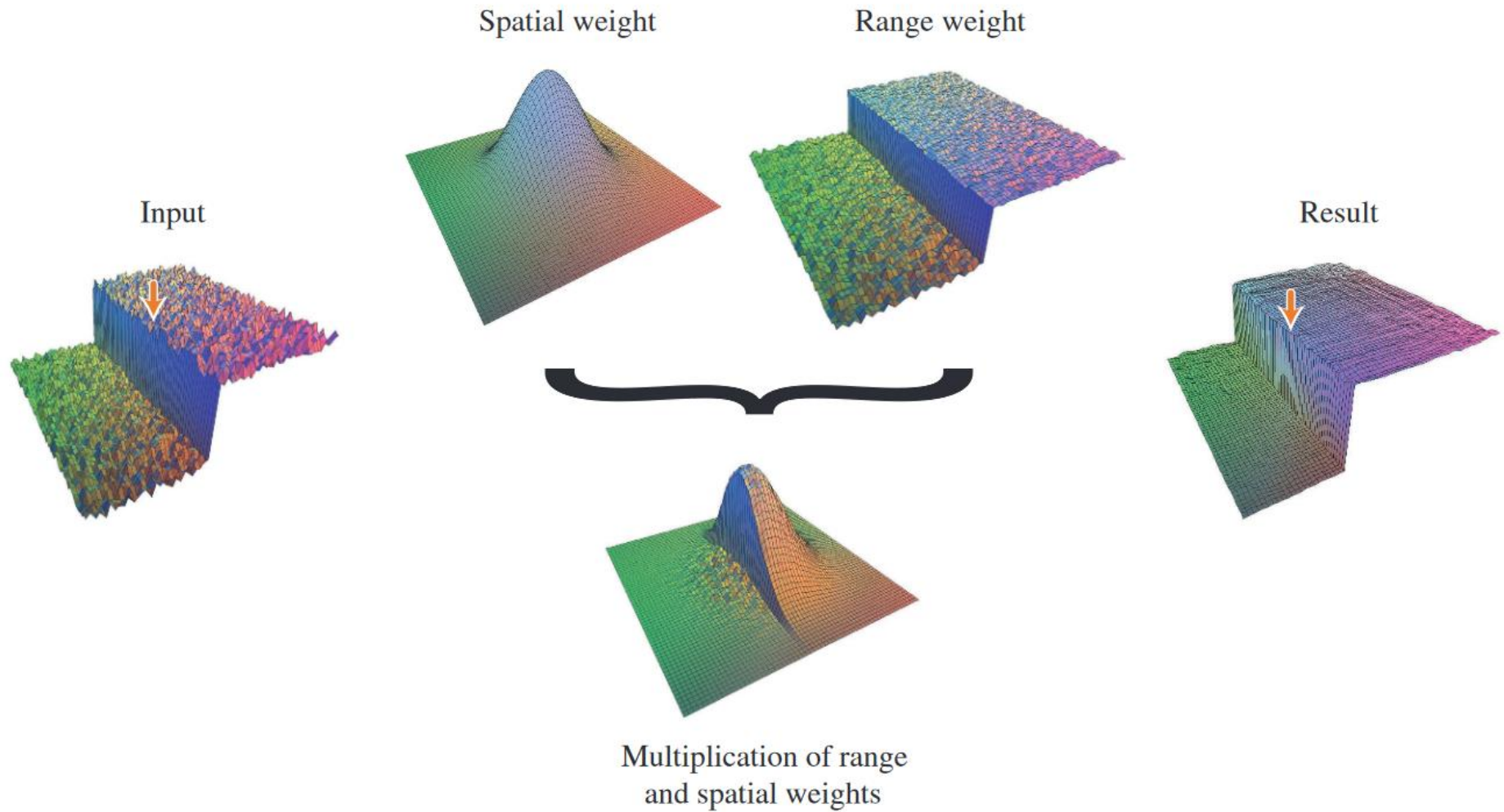
$$BF[I]_p = \overset{\text{new}}{\underbrace{\frac{1}{W_p}}_{\text{normalization factor}}} \sum_{q \in \mathcal{S}} \overset{\text{not new}}{\underbrace{G_{\sigma_s}(\|p - q\|)}_{\text{space weight}}} \overset{\text{new}}{\underbrace{G_{\sigma_r}(\|I_p - I_q\|)}_{\text{range weight}}} I_q$$



space weight



range weight





←
Gaussian
filter

Bilateral filter



- Replace each pixel with the median of its neighbours
- Median M is the value for which the probability that the value in the neighbourhood is less than M is one half.

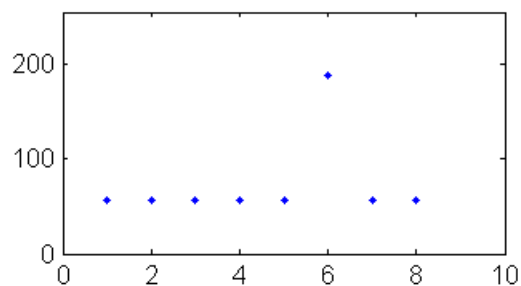
100	98	102
99	105	101
95	100	255

Median: 95 98 99 100 **100** 101 102 105 255

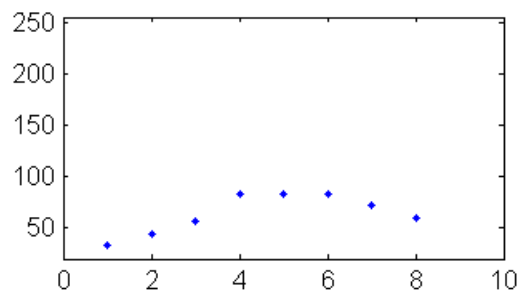
Average = 117.2

- Only partially ordered series suffices for determining the median!
- Typically a neighborhood with an odd number of elements: 3x3, 5x5, etc.

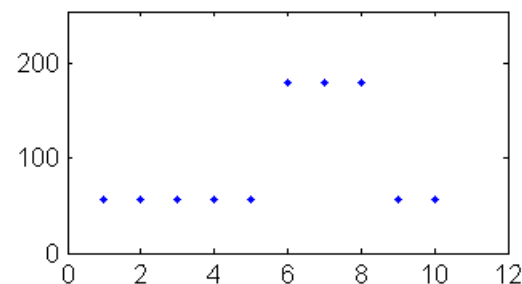
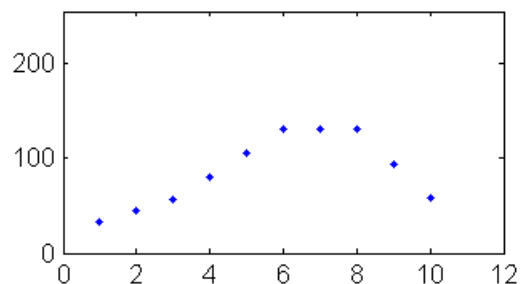
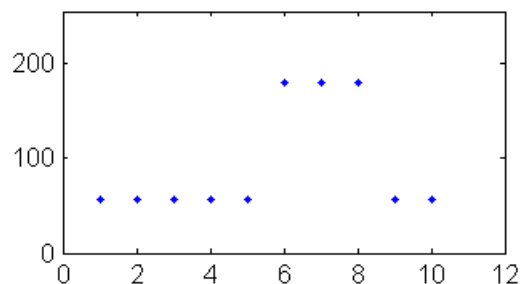
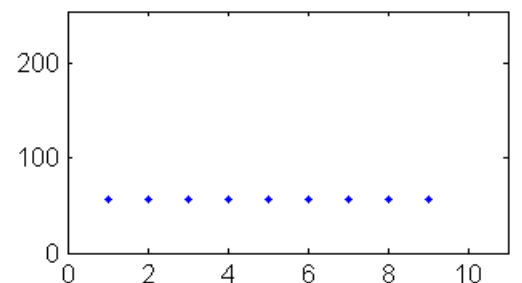
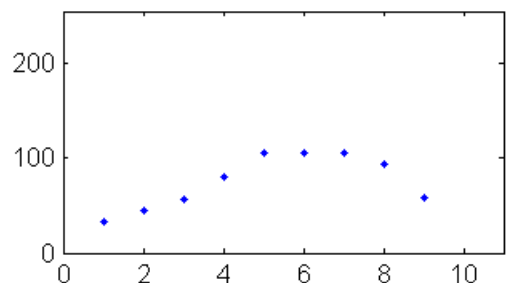
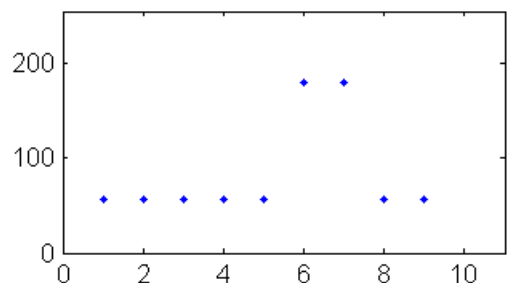
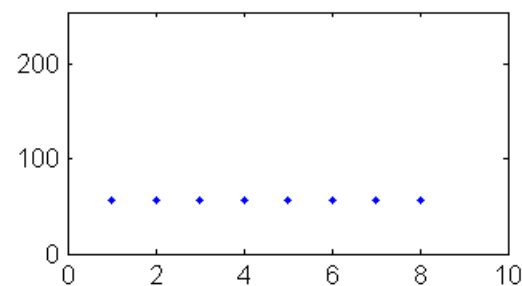
↓ Original



↓ Averaging



↓ Median





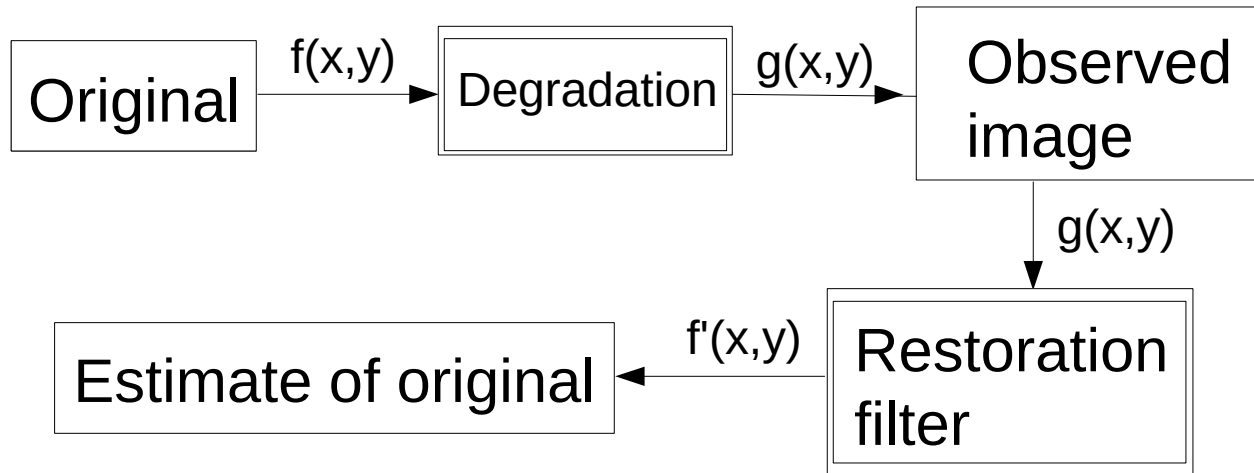
Artif. noise
← Salt-and-pepper
Gaussian →



After median
filtering
← 3x3 5x5 →



- Goal is to improve an image for further processing

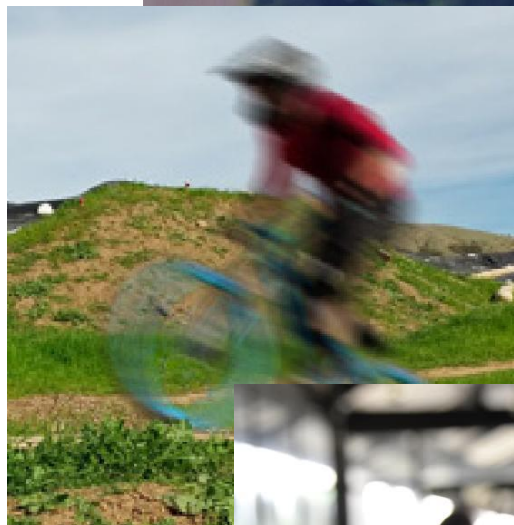


- Generally modelled as a linear degradation function and additive noise term

$$g(x, y) = h(x, y) * f(x, y) + v(x, y)$$

- **Deterministic methods** - known image distortion, low noise
 - Most often, inverse transformation is used to restore the image
- **Statistical methods** – an optimization criterion
 - E.g. Least Squares Method (LSM)
- The more we know about image distortion, the better the image restoration results.

- Optical system defects (lenses)
- Nonlinearity of optical sensors
- Relative movement between object and camera
- Wrong focus
- Atmospheric turbulence (uneven temperatures in the atmosphere cause bending of light rays) - astronomy



- **Mathematical modeling** – Impulse response is described analytically
 - Camera motion, optical diffraction, etc.
- **Estimating by experimentation**

- A star very far

→
PSF of Hubble
telescope with faulty
optics



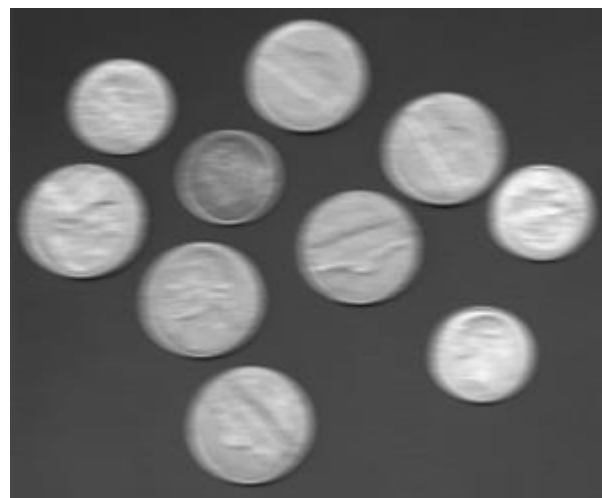
- **Identification from distorted images** – patterns with lines and edges
 - Line Spread Function (LSF)
 - Step Spread Function (SSF)

- Object moves at a constant speed V along the x -axis relative to the camera at the time T the shutter is opened
- The analytical solution of the degradation function is

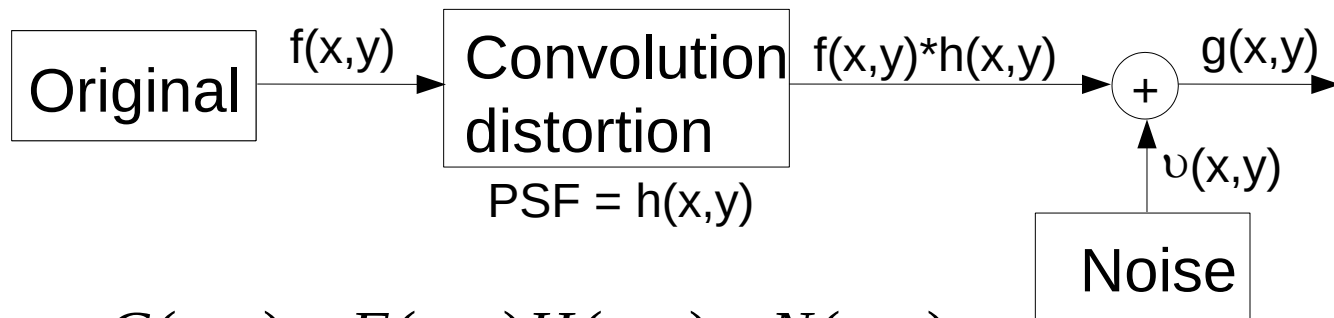
$$H(u, v) = \frac{\sin(\pi VTu)}{\pi Vu}$$



Artificially
generated
motion
blur
→



- Inverse convolution – deconvolution
- Linear image distortion model



$$G(u, v) = F(u, v)H(u, v) + N(u, v)$$

- Distortion is known – $h(x, y)$ or $H(u, v)$
- Noise $v(x, y)$ is independent
- Inverse filter

$$M(u, v) = H^{-1}(u, v) = \frac{1}{H(u, v)}$$

$$F'(u, v) = \frac{G(u, v)}{H(u, v)} = F(u, v) + \underbrace{\frac{N(u, v)}{H(u, v)}}_{\text{Error term}}$$

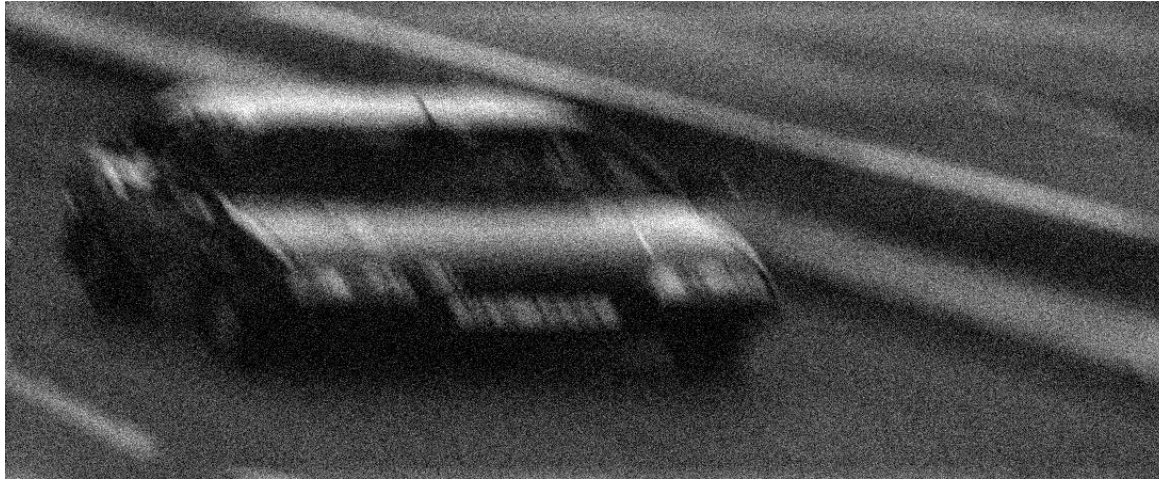
- The additive error term has a large effect on the frequencies where $H(u, v)$ has a small value (usually high frequency $\rightarrow$ image blur)
- **Can only be used if noise $\nu(x, y)$ is minimal!**
- Frequency range limitation $\rightarrow$ pseudoinverse filtration



Image blurred by
movement
between object
and the camera

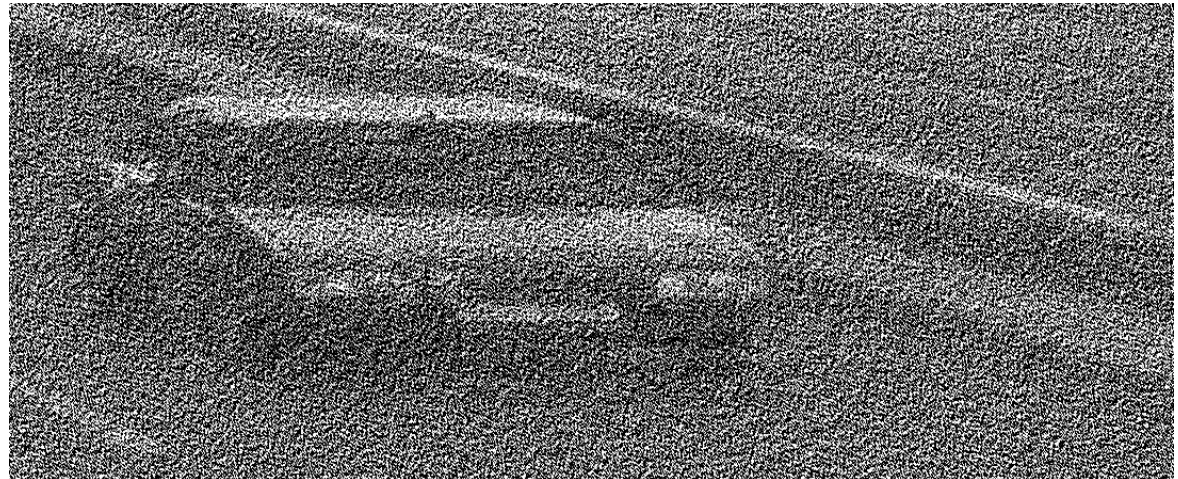
→
Result of the
inverse filtering





Motion blurred
image + artificially
added Gaussian
noise

→
Result of the
inverse filtering



- We have the degraded image $g(x,y)$
- We are looking for an estimate of the original image $f'(x,y)$ to satisfy a given criterion
- We know various characteristics of noise and distortion
- The different methods differ in what we know.
- Wiener filtr, see further slides
- ... deconvolution – total noise energy, etc.
- Blind deconvolution – we know nothing

- Best known improvement to *inverse filtering*
- It works in presence of noise with known statistical properties
- Noise is signal-independent
- Seeks an estimate f' that minimises the statistical error function

$$E\{(f(i, j) - f'(i, j))^2\} \rightarrow Min$$

where E is the mean square error

- The solution is

$$M(u, v) = \underbrace{\frac{1}{H(u, v)}}_{\text{Simple inverse filter}} \underbrace{\frac{|H(u, v)|^2}{|H(u, v)|^2 + \frac{S_{vv}(u, v)}{S_{ff}(u, v)}}}_{\text{Wiener's correction factor}}$$

- Wiener correction factor
 - Values in the range $\langle 0, 1 \rangle$
 - $S_{vv}(u, v)/S_{ff}(u, v) \rightarrow 0$... low noise-to-signal power ratio
WKF $\rightarrow 1$
 - $H(u, v) \rightarrow 0$... due to distortion some frequency has lost, so it is not worth amplifying noise
WKF $\rightarrow 0$

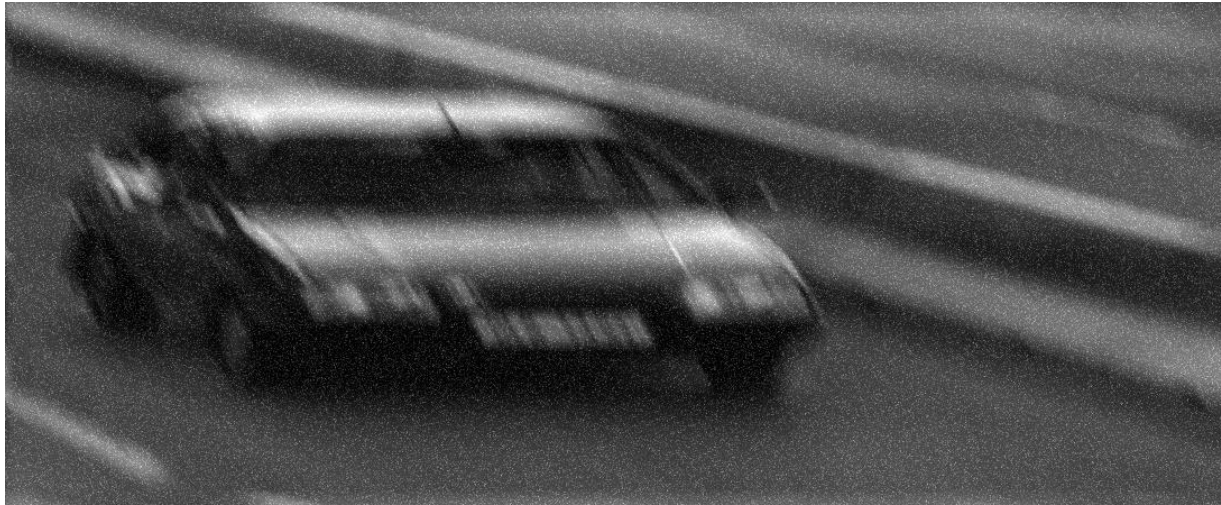
- The solution is

$$M(u, v) = \underbrace{\frac{1}{H(u, v)}}_{\text{Simple inverse filter}} \underbrace{\frac{|H(u, v)|^2}{|H(u, v)|^2 + \frac{S_{vv}(u, v)}{S_{ff}(u, v)}}}_{\text{Wiener's correction factor}}$$

- What is wrong here?

- Modified version – only signals coming out of the system
 - No original image required

$$M(u, v) = \frac{1}{H(u, v)} \frac{S_{gg}(u, v) - S_{vv}(u, v)}{S_{gg}(u, v)}$$



Motion blurred
image + artificially
added Gaussian
noise



After Wiener
filtering

